

CHENG CHENG

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Education

Indiana University Bloomington

Ph.D. in Biostatistics

Aug. 2026 (Expected)

Bloomington, IN, USA

- Expected research focus: data-driven and Bayesian modeling of infectious diseases
- **Research Interests:** Computational Epidemiology, Infectious Disease Modelling

University of Malaya

Master of Science in Mathematics and Statistics

Mar. 2024 - Oct. 2025

Kuala Lumpur, Malaysia

- Data-driven modeling and analysis of infectious diseases
- **Research Interests:** Computational Epidemiology, Deep Learning, Bayesian Inference

Yuncheng University

Bachelor of Science in Mathematics and Applied Mathematics

Sep. 2019 - May. 2023

Yuncheng City, China

- **Relevant Coursework:** Mathematical Modeling (97%) Mathematical Analysis II(93%) Advanced Algebra(91%)

Projects

Dynamics-Informed Neural Networks (DINNs) for COVID-19 Forecasting

Dec. 2023 – Jan. 2025

- Integrated a SEIRV compartmental model into a deep neural network, enforcing both data fit and ODE residuals.
- Learned time-varying transmission rate $\beta(t)$ and vaccination rate $\epsilon(t)$, and quantified prediction uncertainty via an ensemble of models.

Enhancing Cardiac Electrophysiology Modeling: Arrhythmia Prediction & Drug Safety

Feb. 2025 – Jul. 2025

- Developed a parallel computing framework for large-scale parameter sweeps on a rabbit ventricular action potential model.
- Incorporated ion-channel block dynamics to assess anti-arrhythmic drug safety and improved model accuracy in predicting proarrhythmic risk.

Publications

Cheng Cheng, Elayaraja Aruchunan, Muhamad Hifzhudin Noor Aziz*. (2025). Leveraging dynamics informed neural networks for predictive modeling of COVID-19 spread: A hybrid SEIRV-DNNs approach. *Scientific Reports*, 15(1), 2043. doi: 10.1038/s41598-025-85440-1

Chunjuan Zhu†, **Cheng Cheng**†, Jing Guo, Xiaomei Feng*, Qiang Sun*. (2025). Retrospective analysis of estimated serial interval and reproductive number of SARS-CoV-2 Omicron variant in Korea. *Advances in Continuous and Discrete Models*, 2025(1), 43. doi: 10.1186/s13662-025-03906-y

Rui Hu, Elayaraja Aruchunan*, Muhamad Hifzhudin Noor Aziz, **Cheng Cheng**, Benchawan Wiwatanapataphee. Dynamic analysis and optimal control of a fractional-order epidemic model with nucleic acid detection and individual protective awareness: A Malaysian case study. *AIMS Mathematics*. doi: 10.3934/math.2025724

Xiaomei Feng, Yuan Chen, **Cheng Cheng**, Lei Wang *. The Dynamics of Hepatitis B transmission with treatment and vaccination: A case study in the Ili Region of Xinjiang, China by A Physics-Informed Neural Networks Approach. *Mathematical Biosciences* . 2026

Sagar Kumar, **Cheng Cheng**, Mikkel Nagorsen, Athira Satheesh Kumar, Sukanya Krishna, Anne Bischops, Marie-Laure Charpignon, Maiamuna S. Majumder*. *Submitted*.

Rui Hu†, **Cheng Cheng**†, Elayaraja Aruchunan*. Application of a media influence integrated physics-informed neural networks based SEIHR model for COVID-19 dynamics in Malaysia. *Ain Shams Engineering Journal (Under Review)*.

Cheng Cheng, Rui Hu, Xiaomei Feng*. Modeling dengue transmission dynamics in Guangdong Province: A mosquito-mediated model with population mobility. *working paper*.

Awards

University of Malaya Scholarship	02/2025
Nomination for the Mathematical Modeling Competition in Shanxi Province	04/2022
National Third Prize in the National Student Market Research and Analysis Competition	05/2022
National Second Prize in the National Student Mathematical Modeling Competition	11/2021

Internships

Research Volunteer	Dec 2024 – Present
Plan B/ Hurricane Evacuations	Computational Epidemiology DVRN
Research Assistant	Feb 2025 – July 2025
Enhancing cardiac electrophysiology model: Improving arrhythmia prediction and drug safety evaluation	University of Malaya

Skills

Software: Python, Matlab, R, Latex, Julia, ArcGIS
Language: Chinese(native), English(fluent)